Judes Tumuhairwe

Staff Software Engineer | Distributed Systems

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Professional Summary

■ Tech Lead engineering Tier-0 internet-scale distributed systems, capacity efficiency, and infrastructure optimization for fault-tolerant, low-latency platforms. Brings a language-agnostic approach to rapidly mastering and deploying new technology stacks to optimize Critical User Journeys.
■ A force multiplier with a passion for mentoring junior engineers, sets examples of good code and best practices and helps other ICs contribute to their full potential
■ Sole committer and maintainer of (Experiment4J) passionate about mentoring teams, setting high architectural bars, and partner-aligning across engineering, legal & compliance and product.
■ tumuhairwe.github.io

Experience

Senior Software Engineer — Aug 2024 – Present

BILL — San Jose, CA

Team: The Invoice Financing (IF) team at BILL is responsible for short-term collateral borrowing where small businesses use outstanding customer invoices to secure immediate cash advances.

Impact: We (the IF team) bring in about $2.4 to $2.6 million every quarter in net new revenue.

My Role

- **System Ownership & Modernization**: Drove a multi-year architectural roadmap to decompose a 15-year-old, revenue-critical monolith into a high-throughput microservices architecture utilizing the Strangler Fig pattern, supporting a working capital ecosystem generating $2.4M–$2.6M in net new quarterly revenue.
- **High-Throughput Operations**: Decomposed revenue-critical Micronaut and Java 17 backend services integrating high-throughput streams via Kafka and optimied caching tiers via Redis to sustain low-latency SLAs.
- **Observability & Health**: Owned the complete software lifecycle from design through production deployment, building robust telemetry dashboards via Splunk and Tableau to manage low-latency SLAs.
- **Technical Leadership**: Lead technical design discussion, write design docs and enforce a high bar for code quality through rigorous peer reviews.
- **API & Service Design**: Implement business-critical backend functionality utilizing Apollo GraphQL APIs and REST endpoints, routed to a performant Micronaut and Java 17 service tier.

Environment

- Micronaut, Gradle, Oracle, GraphQL, SonarQube, Docker & Kubernetes, Kafka, Redis, AWS Splunk

Tech Lead — Jul 2022 – Nov 2023

Capital One — McLean, VA

- **Team**: As the Tech Lead for the Internal Digital Recoveries(IDR) teams in the Card Tech org I was responsible for leading 3 different vendor teams to bring recoveries in-house (instead of being sold to debt-buyers).

Impact: The IDR project lead to $208 million in net new revenue.

My Role:

- **Distributed Systems & Storage Governance**: Led 3 cross-functional, geographically distributed engineering teams across multiple time zones in the design, deployment, and operational tuning of highly available, fault-tolerant microservices managing high-volume data writes and distributed SQL clusters (Postgres), Kafka, and AWS (RDS, Lambda).

- **Strict Operational Ownership**: Maintained live-service operations on a rotational, on-call basis, leveraging New Relic, Datadog, CloudWatch, and Splunk to achieve rapid real-time incident resolution and maintain uptime postures for high-throughput financial transactions.
- **High-Stakes Technical Governance**: Authored and defended mission-critical technical design papers before the enterprise Design Review Board, ensuring recovery-system architectures adhered to Privacy by Design principles and strict data governance standards.
- **Organizational Leverage & Alignment**: Facilitated strategic alignment between cross-functional squads (Card Tech, Legal, and Finance) to smoothly migrate core recovery infrastructure in-house, replacing legacy external vendor systems with a centralized, scalable platform that generated $208M in net new revenue.

Environment
Spring Boot, Gradle, Postgres, Angular, SonarQube, Datadog, Docker & Kubernetes, Kafka, MongoDB, SAFe, AWS (EC2, RDS, Lambda, CloudWatch), New Relic, Splunk, Snowflake, Java 21

Senior Software Engineer — Oct 2019 – Apr 2022
Ford Motor Company — Dearborn, MI

- Architected and operated the Connected Vehicle Data Operating System (CVDOS), a core internal developer platform that enables Ford engineers to build, test and deploy vehicle commands at scale. We engineered the tool that reduced the ideation-to-prototype lifecycle by 65% for internal engineering customers.

Impact: The CVDOS project reduced time from ideation to prototype by 65% for engineers at Ford.
My Role:

- **Developer Platform Architecture:** Developed cloud-native Spring Boot microservices using Kafka to orchestrate asynchronous message dispatching and response re-aggregation across global vehicle fleets, leading to a 65% reduction in ideation time.
- **SDK Integration & Data Ingestion:** Led integration of the Microsoft SDK to consume high-throughput telemetry and command responses from vehicles transforming raw signal into actionable datasets for internal customers.
- **Engineering Velocity & Mentorship**: Created foundational platform onboarding documentation and directly mentored junior engineers on optimizing codebases for non-functional requirements, including system latency, scalability, and code maintainability.
- **Cross-Team Influence & Blameless Culture**: Fostered a rigorous culture of system reliability by spearheading cross-functional code reviews and leading blameless post-incident retrospectives to maintain strict enterprise quality standards.

Environment
Java 17, Spring Boot, SQL Server, Angular, Splunk, Kafka, SonarQube, Redis, Pivotal Cloud Foundry, Google Cloud Platform, PostMan, Git

Software Engineer — Apr 2017 – Sep 2019
Cengage Learning — Farmington Hills, MI

- **My team:** We were responsible for building the subscription platform ("Cengage Unlimited") to counter dwindling textbook sales by making thousands of books at the "Netflix of publishing" available in various formats.

Impact: Cengage Unlimited was the first in the country to introduce a subscription-based model to publishing which directly increased recurring revenue.
My Role:

- **High-Scale Platform Integration**: Architected core Identity and Access Management (IAM) abstractions for a subscription platform supporting 1.5M+ active users, optimizing first-time user onboarding flows, conversion funnels, and retention hooks..

- **Fault-Tolerant Architecture**: Designed resilient, non-blocking execution flows using RxJava and Hystrix circuit-breakers to mitigate systemic failures and protect system uptime during high-traffic peaks.
- **Technical & Agile Leadership**: Mentored junior software engineers in adopting production best practices, championing the integration of comprehensive unit and integration test suites, automated CI/CD workflows, and regular post-mortems.
- **Full-Stack Collaboration**: Partnered closely with UX/UI designers and product owners to scale a unified student dashboard, securely caching and serving thousands of concurrent, personalized content recommendations (eBooks, video, and audio streams).

Environment
Java 8, Spring Boot, RxJava, Hystrix, Maven, Bower, Splunk, Jenkins

Software Engineer — Feb 2016 – Apr 2017

Advaita Bioinformatics — Plymouth, MI

My team was responsible for building the tools used scrape data from public repositories, parse it and upload it into the search index.
Impact: We created the 1st one-stop shop for genomic data in the bioinformatics industry
My Role:
■ I developed cloud native microservices, writing scripts to collect, parse and upload data from multiple public repositories and improving the user-experience.
■ I developed and integrated UI controls with multiple java microservices.
■ I wrote bash and perl scripts to download, parse and merge data from multiple public repositories and integrate it into our search engine index and other applications.

- **Environment**
  Agile, Spring Boot, Hibernate, Zuul, Eureka, S3, SNS, SQS, RDS

Software Engineer — Sep 2014 – Dec 2015

Ithaka/JSTOR — Ann Arbor, MI

Team: My team ("Content Ingest") was responsible for receiving content (zip files) from publishers of books and journals and itemizing, characterizing, and preparing them for use by other teams (e.g. search-index team) and to build the UI publishers use to upload and track new journals/issues as they move through different states
Impact: With Content Ingest publishers can quickly upload new editions of journals and make them searchable by all universities, which increased revenue for both Ithaka and the publishers

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My Role:

- **Distributed Storage Ingestion:** I was responsible for monitoring high-throughput data ingestion pipelines and optimizing distributed storage layout using Cassandra and Apache Solr to itemize characterize and index massive unstructured, datasets from global publishers.
- **Service Discovery & Monolith Modernization:** Proposed and implemented code and architectural changes to move ingestion processing away from Hadoop to REST API. Enabled service-location using Eureka.
- **State Machine Implementation:** Implemented tracking of content as it went through different phases using finite state machine
- **Operational Readiness**: Participated in PagerDuty rotation to help customers understand issues, triage and often resolve them.

Environment
Spring, Cassandra, Hadoop, Swagger, Amazon S3, SNS, SQS, Apache Solr

Software Engineer — Aug 2012 – Sep 2014

Domino's Pizza — Ann Arbor, MI

Team: My team ("Platform team") built and scaled the global ordering platform backend, ensuring 99.99% availability for a multi-client ecosystem (Web, iOS, Android) during peak global traffic events.

Impact: Scaled the Platform so it could support more front-ends, leading to more orders and increased revenue e.g. serving millions of concurrent users during Superbowl
My Role

- **Extreme Scale Caching**: Performance-tuned core global ordering platform infrastructure to support millions of concurrent users during peak traffic events (Super Bowl), architecting multi-level caching strategies (EhCache) to maintain 99.99% availability and ultralow latency.
- **Global Architecture & Internationalization**: Spearheaded the data-storage internationalization and localization of backend systems to dynamically support rapid global expansion and regional data standards.
- **Latency Optimization & Resilience**: Implemented multi-level caching strategies using EhCache and utilized Gatling/JMeter for rigorous load testing to identify and eliminate system bottlenecks, ensuring a low-latency, personalized ordering experience.
- **System Decoupling & Orchestration:** Decomposed monolithic ordering logic into resilient microservices and integrated Mule Enterprise Service Bus (ESB) for sophisticated service orchestration, message routing, and data transformation.
- **Fault-Tolerant Design:** Engineered localized menu and pricing modules to ensure system reliability across multiple locales, maintaining a consistent high-availability posture during complex global rollouts.

Environment
Spring, JMeter, Jenkins, SoapUI, Oracle, Mule ESB, Cucumber, Gatling, MuleSoft

Software Engineer — Jan 2008 – Jul 2012

Ford Motor Company — Dearborn, MI

- Project 1:
- Part Library 2 is a repository of all the parts used on all the vehicle-programs in Ford PL 2 eased recalls and has saved the company $7 million annually data integrity gave the designers and business a single view of the data

My Role

- ■ Managed the 'journey' of millions of data records through custom ETL processes, ensuring data integrity and building robust ingestion gatekeepers for high-volume upstream sources.
  ■ Designed and built batch jobs to periodically fetch data from upstream external systems
  ■ Interacted with vendor APIs to extract data, transform it based on dynamic logic, and load (or ignore) it into the new system. Also wrote the UI for users to request ad-hoc batch jobs
  ■ Designed a notification system (email, pager) to be invoked when certain data becomes available

Project 2
Automated Certification System (ACS) is a system to automate the certification of new engines by the EPA. Before, tests and certification took a year but with ACS, it now takes 3 months enabling faster turnaround, improvement in failing designs and faster time to market.

- My Role

Modernized core data infrastructure by migrating legacy systems to high-efficiency Java services designed for automated, large-scale data processing and validation.
Project 3:

- Cost Reduction Idea Database (CRID2) is a system to automate the collection of ideas from engineers, suppliers, business managers, and partners on how to reduce costs anywhere in the production chain (from conceptualization, design, production to advertising and marketing) across all vehicle programs structured way that can be quickly evaluated, rated, approved and applied.
- Project 3:
  ■ Implemented a finite state machine to process business logic as data goes from submitted to approved/integrated for various models
  ■ Designed the data model in concert with the data architect
- **Environment**
  Agile, J2EE, IBM Websphere, Oracle 10g®, MS SQL Server, IBM Message Queue®

Education

Master of Science (M.S.) in Applied Information Systems — May 2006
Towson University — Towson, MD, US

Bachelor of Science (B.S.) in Computer Information Systems — May 2004
Towson University — Towson, MD, US

Bachelor of Science (B.S.) in Business Administration — May 2004
Towson University — Towson, MD, US

Skills

- **Programming Languages**: Java (Expert), Go (Language-agnostic/Ramping), Python (Language-agnostic/Ramping)
- **Storage & Caching**: Redis, Distributed SQL (Postgres), Cassandra, MySQL, Memcached, Oracle, MongoDB, EhCache
- **Infrastructure & Streaming**: Kafka, Docker, Kubernetes, AWS (RDS, S3, Lambda, EC2), Cloud-Native Systems
- **AI & Productivity Tools:** GitHub Copilot, Gemini, n8n, GitLab Duo
- **Technical**: Docker & Kubernetes, Spring, Kafka, SQL, NoSQL & Databases
- **Other**: Tech Debt Management, API Design, Mentoring and coaching

Certifications

- AWS Certified Solutions Architect
- AWS Certified Developer

Publications

- Cloud computing is not just for Fortune 500, 06/02/2023, Judes Tumuhairwe, https://minoritybusinessreview.com/cloud-computing-is-not- just-for-fortune-500

Open Source Leadership

Experiment4J (Sole Maintainer & Committer)

- **Ecosystem Stewardship**: Forked and revitalized a refactoring framework, a resilient developer library enabling safe production A/B testing, feature flagging, and rigorous runtime experimentation flows of critical paths.
- **Safe Code Experimentation**: Engineered asynchronous execution abstractions and API hooks that allow internal/external engineering teams to validate candidate code paths with zero user-facing latency degradation.
- **Paper-Driven Feature Evolution**: Actively spearheading the project roadmap by implementing core testing and isolation features adopted from GitHub's original Ruby implementation papers for the Scientist pattern.
- **Telemetry Integration**: Built automated real-time performance tracking and telemetry reporting directly into InfluxDB to safely analyze execution behaviors during massive architectural shifts

Databases

Oracle, MySQL, SQL Server, Postgres
Snowflake, Cassandra
Redshift, DynamoDB, Redis, MongoDB